Specification No 322

XLPE Power Cable

322-1 **General.** This Specification covers the general requirements for design, manufacture, test and supply of XLPE power cable.

The specific ratings, characteristics and the special requirements and features of the equipment not covered herein are given in the accompanying Ratings and Features sheet (if any).

322-2 **Materials and Workmanship.** All materials shall be new and shall be the best available for the purposes used, considering strength, ductility, durability and suitability for the intended services and best engineering practice. Workmanship shall be of the highest grade and in accordance with the best modern standard practice.

322-3 **Service Conditions.** All materials shall be suitable for installation and use at an altitude of 1000 m or less in a tropical climate with a maximum ambient temperature of 45°C and 100% relative humidity for outdoor equipment (or a maximum ambient temperature of 40°C and 90% relative humidity for indoor equipment) without corrosion, deterioration or degradation of performance characteristics.

322-4 **Codes and Standards.** All equipment, materials, devices, fabrication and testing shall conform to the codes, specifications and standards listed below and all applicable codes, specifications and standard referred therein.

ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
ICEA	Insulated Cable Engineer Association
IEC	International Electrotechnical Commission
NEMA	National Electrical Manufacturers Association
TIS	Thai Industrial Standards Institute
CIGRE	***Conseil International des Grands Réseaux Électriques***

All threaded parts requiring external connection shall have UNC screw and pipe threads. All internal parts may have threads in accordance with the established specification in the country of manufacturer.

It is the intent that all equipment, materials, devices, fabrication and testing shall conform to the application codes, specifications and standards even though they are not specifically noted herein. Equivalent codes, specifications and standards established and approved in the country of equipment or material manufacture may be used subject to EGAT's approval. If this election is made, the Bidder shall so state and include in his bid the governing codes, specifications, and standards proposed together with an itemized list of specific deviations from the requirements of codes, specifications and standards referred herein.

The latest issue of all codes, specifications and standards shall govern.

The most stringent requirement, in the event of code, specification or standard conflict, shall govern. This specification shall govern in the event of discrepancies between it and applicable codes, specifications and standards.

322-5 **Design and Construction.** The general construction of the cable shall be as shown on drawings being furnished with the characteristics described herein.

322-5.1 33 kV and below XLPE Power Cable.

a. **Conductor.** The conductors shall be plain annealed copper compacted round concentric lay stranded construction conformable to the applicable requirements of IEC Publication No 60228 Class 2 or compact segmental stranded.

b. **Conductor Shield *(Conductor Screen)*.** The conductor shall be shielded with an extruded layer of thermosetting semi-conducting material or the combination of helically wrapped semi-conducting tape over the conductor and then an extruded layer of thermosetting semi-conducting material. In either case, the extruded layer shall be firmly bonded to the insulation.

c. **Insulation.** The insulation shall be cross-linked polyethylene conforming to the requirement of ICEA Pub. No S-93-639.

   The insulation shall be capable of operating continuously at conductor temperatures not exceeding 90°C for normal operation, 130°C for emergency overload conditions, and 250°C for short circuit conditions.

   The cable shall be provided with a 133% insulation level.

d. **Insulation Shield *(Insulation Screen)*.** The insulation shall be shield with a layer of semi-conducting material applied directly over the insulation.

e. **Metal Shield *(Metal Screen/Sheath)*.** The metal shield shall be annealed copper tape of suitable width and shall be helically applied over the insulation shielding with minimum 10% lap.

   The annealed copper tape shall be at least 0.1 mm thickness.

f. **Cushion.** A polyester non-woven tape with 10% minimum lap of the tape width shall be wrapped over the metal shield as a cushion layer, the minimum thickness of tape shall not be less than 0.03 mm.

g. **Jacket *(Oversheath).*** The jacket shall consist of a polyvinyl chloride compound conforming to the requirement of ICEA Pub. No S-93-639.

322-5.2 115 and 69 kV XLPE Power Cable.

The XLPE power cable shall be suitable for use in wet or dry location and shall conform to the applicable requirements of TIS 2202 ***and shall be flame retardant according to IEC 60332.***

a. **Conductor.** The conductors shall be plain annealed copper compacted round concentric lay stranded construction conformable to the applicable requirements of IEC Publication No 60228 Class 2 or compact segmental stranded.

b. **Conductor Shield *(Conductor Screen).*** The conductor shall be shield with the combination of helically wrapped semi-conducting tape over the conductor and then an extruded layer of thermosetting semi-conducting material. The extruded layer shall be firmly bonded to the insulation.

c. **Insulation.** The insulation shall be cross-linked polyethylene compound (XLPE) and shall be extruded in tandem process over the extruded semi-conductive conductor shield layer by the Radiant Curing Process (RCP) or suitable completely-dried curing process, so that micro-voids in the insulation are eliminated.

d. **Insulation Shield *(Insulation Screen).*** Insulation shield shall consist of extruded semi-conductive compound which shall be bonded firmly and continuously to the insulation for the requirement of removal before terminating and splicing.

e. **Synthetic Water Blocking Layer.** A semi-conductive non-biodegradable water blocking layer shall be provided under the metal shield to provide a continuous longitudinal watertight barrier throughout the cable length. This layer shall be compatible with other cable materials and shall not affect corroding acting on the metal shield.

f. **Metal Shield *(Metal Screen/Sheath).*** The metal shield shall be a concentric layer of copper wire which is electrically continuous and bonded together throughout the cable length with copper contact tape.

g. Synthetic Water Blocking and Cushion Tape. A non-conductive non-biodegradable water blocking tape shall be applied over the metal shield to provide a continuous longitudinal watertight barrier throughout the cable length. The tape shall have sufficient thickness to perform well as a thermal stress relief layer and to provide for cushion and bedding. The tape shall be compatible with other cable materials and shall not affect corroding acting on adjacent metal layers.

h. Radial Water Barrier. The radial water barrier shall be provided under the jacket. This layer shall consist of aluminum tape coated on both sides with an polyethylene or ethylene acrylic acid adhesive co-polymer.

i. Jacket *(Oversheath)*. The jacket shall consist of extruded black polyvinyl chloride ST_2 (PVC ST_2). Semi-conductive black carbon coating shall be applied on the outer surface of the PVC outer jacket.

322-5.3 230 kV XLPE Power Cable

The XLPE power cable shall be suitable for use in wet or dry location and shall conform to the applicable requirements of IEC Publication No 62067.

a. Conductor. The conductors shall be plain annealed copper compacted round concentric lay stranded construction conformable to the applicable requirements of IEC Publication No 60228 Class 2 or compact segmental stranded.

b. Conductor Shield *(Conductor Screen)*. The conductor shall be shield with the combination of helically wrapped semi-conducting tape over the conductor and then an extruded layer of thermosetting semi-conducting material. The extruded layer shall be firmly bonded to the insulation.

c. Insulation. The insulation shall be cross-linked polyethylene compound (XLPE) ***and shall be*** extruded in tandem process over the extruded semi-conductive conductor shield layer by the Radiant Curing Process (RCP) or suitable completely-dried curing process, so that micro-voids in the insulation are eliminated.

d. Insulation Shield *(Insulation Screen)*. Insulation shield shall consist of extruded semi-conductive compound which shall be bonded firmly and continuously to the insulation for the requirement of removal before terminating and splicing.

e. Synthetic Water Blocking Layer. A semi-conductive non-biodegradable water blocking layer shall be provided under the metal shield to provide a continuous longitudinal watertight barrier throughout the cable length. This layer shall be compatible with other cable materials and shall not affect corroding acting on the metal shield.

f. Metal Shield *(Metal Screen/Sheath)*. The metal shield ***shall be aluminum corrugated sheath covered with Bituminous compound as anti-corrosion layer.***

g. Jacket *(Oversheath)*. The jacket shall consist of extruded black polyvinyl chloride ST_2 (PVC ST_2). Semi-conductive black carbon coating shall be applied on the outer surface of the PVC outer jacket.

322-6 **Cable Marking.** The following items of marking shall be printed on the surface of the outer jacket***/sheath***.

(a) Manufacturer's name or its abbreviation and country of origin.

(b) Size of cable and number of cores

(c) Type of conductor

(d) Type of insulation (e.g. XLPE) and jacket/oversheath (e.g. PVC)

(e) Voltage classification

(f) Year of manufacture

(g) Continuous marking for each meter of the cable length starting from zero through the whole length of cable in each reel.

322-7 **Packing.** Immediately after work of factory test, both ends of power cable shall be greased by silicone paste and covered by PVC end cap, moreover the outer end of power cable above 33 kV through 230 kV shall be ready attached with a pulling eye of sufficient strength having an effective moisture sealing.

Each length of finished cable shall be wound on a suitable non-returnable reel and packed in such a manner which prevents damage during transportation and handling operation. Owing to transportation limitation, the size of reel shall match the transportation limit of not exceeding 3.5 m x 10.0 m x 4.0 m (width x length x height).

The length of the cable to be packed on each reel shall be in accordance with EGAT's requirement. The following information shall be displayed on the side of reel:

(a) Manufacturer's name or its abbreviation

(b) Size of cable and number of cores

(c) Type of conductor

(d) Type of insulation (e.g. XLPE) and jacket/oversheath (e.g. PVC)

(e) Voltage classification

(f) Year of manufacture

(g) Rolling direction of reel and terminal position of cable

(h) Length of cable

(i) Reel number and other suitable identification for reel and reel size

(j) Net weight and total weight (kg.)

(k) Item No.

(l) Contract No.

Additionally, the information in (h), (i), (j), (k) and (l) shall be highlighted in contrasting colors for clearly visual inspection.

322-8 **Test.** Test of equipment shall be performed according to the requirements specified in each referred test item. The costs of all tests and reports shall be borne by the Contractor.

322-8.1 **Design Test *(Type Test)*.** The proposed equipment shall already have the design test record of the same type and same rating as EGAT called for, in accordance with the tests specified in each equipment test items, which is subject to EGAT′s approval (unless otherwise specified in each equipment test item). The design test record shall be required and submitted together with the tender document during the bidding.

322-8.2 **Routine Test.** Each equipment shall be completely assembled at the factory and subject to the tests specified in each equipment test item.

322-8.3 **Quality Conformance Test *(Sample Test)*.** The quality conformance test shall be performed by selecting the sample from each lot of equipment. The numbers of samples for the test shall be as specified in the Standard.

322-8.4 ***Prequalification Test (PQ Test). Prequalification test shall be conducted prior to the commercial supply of the proposed Cable System (proposed Cable and Cable Termination which are installed and tested together in the same test loop), in order to demonstrate the satisfactory long-term performance of the complete cable system. The PQ test shall be submitted together with tender document during the bidding.***

322-8.5 ***Extension of prequalification Test (EQ Test). Extension of prequalification test shall be conducted prior to the commercial supply of the proposed Cable System, in order to demonstrate satisfactory long-term performance of the complete cable system, taking into account an already prequalified cable system. The EQ test shall be submitted together with tender document during the bidding.***

322-8.6 **Test Report.** The report of all tests, curves and standard application data shall be furnished to EGAT immediately after completion of the tests.

322-8.7 **Test Procedure.** The Contractor shall submit the test procedure of specified tests and actual design tests to EGAT for approval. The test procedure shall consist of procedures, applied voltage, current and criteria to justify the result of the tests.

322-8.8 **Test Items for 33 kV and below XLPE Power Cable.** The test shall be performed in accordance with the latest ICEA Pub. No S-93-639, ASTM B3, ASTM B8.

a. **Design Tests.** These tests may be omitted if a design test record can be submitted.

(a) **Conductor Tests.** Tests shall be performed on selected samples of the conductor before the application of any covering.

1. Tensile strength test.

2. Elongation test.

3. Conductor resistivity tests.

4. Dimension measurements.

5. Surface finish inspection.

(b) **Physical and Aging Tests on the Cables, Insulation and Jacket.** Tests shall be performed on selected samples of the cable insulation and jackets.

1. Thickness measurement.

2. Tensile strength test.

3. Elongation test.

4. Aging test.

5. Heat distortion test.

6. Capacitance and power factor tests.

7. Accelerated water absorption.

8. Jacket heat shock test.

9. Jacket heat distortion test.

10. Resistivity test.

11. Flame resistance test.

b. **Routine Tests.** Electrical tests shall be performed on the entire length of completed cables.

1. AC voltage test.

2. Insulation resistance test.

3. Partial discharge extinction level test.

322-8.9 **Test Items for 115 and 69 kV XLPE Power Cable.** The tests shall be performed in accordance with the latest TIS 2202 ***and IEC 60332.***

a. **Design Tests.** These tests may be omitted if a design test record can be submitted.

1. Bending followed by partial discharge.

2. Tan δ measurement.

3. Heating cycle voltage test.

4. Partial discharge test.

5. Lightning impulse voltage test followed by a power frequency voltage test.

6. Resistivity of semi-conducting layers.

7. Check of cable construction.

8. Test for determining the mechanical properties of insulation before and after aging.

9. Test for determining the mechanical properties of non-metallic sheaths before and after aging.

10. Aging tests on pieces of complete cable to check compatibility of materials.

11. Pressure test at high temperature on sheaths.

12. Hot set test for XLPE insulation.

13. Shrinkage test for XLPE insulation.

14. Water penetration test.

15. Heat shock test for PVC ***oversheaths.***

16. Loss of mass test on PVC sheaths.

17. Test under fire conditions. The test under fire conditions in accordance with IEC 60332-1 shall be carried out on a sample of completed cable.

The design test records of the same type and similar ratings of the proposed equipment shall be submitted together with the tender documents during bidding. During the bidding stage, EGAT reserves the right to request the Manufacturer to perform new design test(s) on specified test item(s), in accordance with the agreement between EGAT and the Manufacturer.

The specified test item(s) shall refer to type tests equivalent to design tests under this Specification relevant to IEC 60840, as identified in TB303. The new design test(s), as agreed upon, may be carried out during the project execution phase but shall not affect the project completion date or delivery schedule as specified in the contract.

The test procedure(s) and design test record(s) of the new test(s) shall be submitted to EGAT for approval. The cost of the new design test(s) shall be borne by the Contractor.

b. Routine Tests

1. Partial discharge test.

2. Voltage test.

3. Electrical test on oversheath of the cable.

c. Quality Conformance Tests

1. Conductor examination.

2. Measurement of electrical resistance of conductor ***and of metal screen.***

3. ***Measurement of thickness of insulation and oversheath.***

4. ***Measurement of thickness of metal sheath.***

5. ***Measurement of diameters.***

6. Hot set test for XLPE insulation.

7. Measurement of capacitance.

322-8.10 **Test Items for 230 kV XLPE Power Cable.** The tests shall be performed in accordance with the latest IEC Publication No 62067 ***and CIGRE TB303.***

a. Design Tests. These tests may be omitted if a design test record can be submitted. ***The tests shall be performed in accordance with the conditions and standards mentioned in IEC 62067***

1. Electrical type tests on complete cable systems.

2. Non-electrical type tests on cable and on cable components.

The design test records of the same type and similar ratings of the proposed equipment shall be submitted together with the tender documents during bidding. During the bidding stage, EGAT reserves the right to request the Manufacturer to perform new design test(s) on specified test item(s), in accordance with the agreement between EGAT and the Manufacturer.

The specified test item(s) shall refer to type tests equivalent to design tests under this Specification relevant to IEC 62067, as identified in TB303. The new design test(s), as agreed upon, may be carried out during the project execution phase but shall not affect the project completion date or delivery schedule as specified in the contract.

The test procedure(s) and design test record(s) of the new test(s) shall be submitted to EGAT for approval. The cost of the new design test(s) shall be borne by the Contractor.

b. Routine Tests. ***The tests shall be performed in accordance with the conditions and standards mentioned in IEC 62067.***

c. Quality Conformance Tests. ***The tests shall be performed in accordance with the conditions and standards mentioned in IEC 62067.***

1. Sample tests on cables.

d. PQ Tests. These tests may be omitted if a PQ test record can be submitted. The tests shall be performed in accordance with the conditions and standards mentioned in IEC 62067.

e. EQ Tests. These tests may be omitted if an EQ test record can be submitted. The tests shall be performed in accordance with the conditions and standards mentioned in IEC 62067.

322-9 Drawings and Documents for XLPE Power Cable
(For Supply Project)

Drawings and documents shall at least comprise the followings:

A. Documents

Item	**Description**	**Printed Documents**	**CD-ROM**
1	Manufacturing and delivery schedule	x	-
2	Test procedure (Factory test)	x	-
3	Test schedule (Factory test)	x	-
4	Test report (Factory test)	x	-

322-10 Drawings and Documents for XLPE Power Cable
(For Supply and Construction Project)

Drawings for Approval, Final Drawing, Documents and CD-ROM shall at least comprise the followings:

Item	***Drawings Title***
1	***Cross section including dimensions***
2	***Technical data***